2022.

Q1.

(b). The navigation equation when $\delta = e$) :

$$\underline{a}^e_{ib} = \underline{f}^e_{ib} + \underline{\gamma}^e_{ib} = C^e_b \underline{f}^b_{ib} + \underline{\gamma}^e_{ib}$$

what we want is $\dot{v}^e_{eb}$, then.

$$r^e_{eb} = r^e_{ib} - r^e_{ie} = r^e_{ib}$$

$$\Rightarrow \dot{r}^e_{eb} = v^e_{eb} = \dot{r}^e_{ib}$$

$$\Rightarrow \dot{v}^e_{eb} = \ddot{r}^e_{ib}$$

$$= -(\Omega^e_{ie}\Omega^e_{ie} + \dot{\Omega}^i_{ie})\underline{r}^e_{ib} - 2\Omega^e_{ie}\underline{\dot{r}}^e_{ib} + a^e_{ib}$$

$$\Rightarrow \dot{v}^e_{eb} = C^e_b \underline{f}^b_{ib} + \underline{\gamma}^e_{ib} - \Omega^e_{ie}\Omega^e_{ie}\underline{r}^e_{ib} - 0 - 2\Omega^e_{ie}\underline{\dot{r}}^e_{ib}$$

$$= C^e_b \underline{f}^b_{ib} + g^e_b(\underline{r}^e_{eb}) - 2\Omega^e_{ie}\underline{\dot{r}}^e_{ib}$$

$$\underline{r}^e_{ib} = \underline{r}^e_{ie} + \underline{r}^e_{eb} = \underline{r}^e_{eb} \Rightarrow \underline{\dot{r}}^e_{ib} = \underline{\dot{r}}^e_{eb}$$

$\therefore$ $\dot{v}^e_{eb} =$ ~~~~~~

(C). From (b) :

$$\underline{\dot{v}}^e_{eb} = C^e_b \underline{f}^b_{ib} + g^e_b(\underline{r}^e_{eb}) - 2\Omega^e_{ie}\underline{v}^e_{eb}$$

$$\Rightarrow \underline{f}^b_{ib} = C^b_e(\underline{\dot{v}}^e_{eb} - \underline{g}^e_b(\underline{r}^e_{eb}) + 2\Omega^e_{ie}\underline{v}^e_{eb})$$

$$= \underline{a}^b_{eb} - g^b_b(\underline{r}^e_{eb}) + 2C^b_e\Omega^e_{ie}\underline{v}^e_{eb}$$

$$\underline{a}^b_{eb} = \underbrace{\underline{\dot{v}}^b_{ib} + \Omega^b_{ib}v^b_{ib}}_{\underline{a}^b_{ib}} - \underbrace{C^b_e\Omega^e_{ie}\underline{v}^e_{eb}}_{\underline{a}^b_{ie}}$$

here : $\underline{a}^b_{eb} = \underline{a}^b_{ib} - \underline{a}^b_{ie}$

we have, $\underline{a}^b_{ib} = \underline{\dot{v}}^b_{ib} + \Omega^b_{ib}\underline{v}^b_{ib}$

$$\underline{a}_{ie}^{b} = \dot{\underline{V}}_{ie}^{b} + \Omega_{ib}^{b} \underline{V}_{ie}^{b} = 0 + 0$$

$$\Rightarrow \underline{a}_{eb}^{b} = \underline{a}_{ib}^{b} = \dot{\underline{V}}_{ib}^{b} + \Omega_{ib}^{b} \underline{V}_{ib}^{b}$$

$$\Rightarrow \underline{f}_{ib}^{b} = \dot{\underline{V}}_{ib}^{b} + \Omega_{ib}^{b} \underline{V}_{ib}^{b} - g_{b}^{b}(r_{eb}^{e}) + 2 C_{e}^{b} \Omega_{ie}^{e} \underline{V}_{eb}^{e}$$

Q3.

(b). The state error is:

$$\tilde{\underline{x}} = \underline{x} - \hat{\underline{x}}$$

$$\dot{\hat{\underline{x}}} = \dot{x} + K(\dot{y} - H\dot{x})$$

The error dynamics equation is:

$$\dot{\tilde{\underline{x}}} = \dot{\underline{x}} - \dot{\hat{\underline{x}}}$$

$$= A\underline{x} + B\underline{u} + w -$$